

Day 10 — Final Assessment + Project Share

Module: Nikhilaṃ Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will complete the summative quiz, present their project artefact in 3 minutes, and submit a brief written self-reflection.

Materials

- quizzes/summative.md (printed, one per student)
- Stopwatch for project presentations
- Camera (phone) for documenting projects, if permitted by school policy
- Stickers for completion (optional, but motivating)

Lesson flow

Warm-up (3 min)

- Final sūtra recitation, in unison, from memory — *Nikhilaṃ Navataścaramaṃ Daśataḥ*.
- One sentence each, going around the room: “*The one thing I’ll remember from this module is...*”

Summative quiz (20 min)

- Closed notebook.
- 25 marks. Sections: complement recall, subtraction practice, general subtraction, conceptual short answer, open response on “why it works.” See quizzes/summative.md.
- Strict 15-min time limit on the multiple-choice / short-answer parts; remaining time for the open response.
- Collect.

Project presentations (18 min)

- Each pair presents for 3 minutes (set the timer visibly).
- Format depends on artefact:
 - **Shopkeeper chart:** Pair demos using pretend currency on a real scenario.
 - **Teaching video:** Pair plays the 60-sec video, then takes 90 sec of Q&A.
 - **Two’s-complement poster:** Pair walks through the poster’s argument.
- Rest of class fills in a quick “feedback card” — one strength, one question.

Wrap-up (4 min)

- Quick poll: was Module 9 useful? Surprising? Hard?

- Honest history note one more time: “*You learned a real, useful method. The sūtra is from Tirthaji’s 1965 book. The deeper history is more recent than the name ‘Vedic’ suggests. You can hold both: the method is real; the historical claim is unverified.*”
- Self-reflection slip (1 minute): on the back of the quiz, complete:
 - One thing I learned: _____
 - One thing I’m still confused about: _____
 - My favourite Sanskrit term from the module: _____

Student-facing content

Today’s tasks

1. Summative quiz (20 min, closed notebook).
2. Project presentation (3 min, with your partner).
3. Self-reflection (1 min, on the back of the quiz).

You’ve come a long way. Day 1’s 10 000 – 4567 took most of you about 30 seconds. By now you can do it in 5.

Homework

- None. Module ends.

Differentiation

- **Tier up:** If finished early on the quiz, write the algebraic identity for the Nikhilam method in full generality (for any base b).
- **Tier down:** Extra 5 min on the quiz allowed for slow processors; quiet pull-aside support.

Teacher notes

- The summative is the most marked single assessment of the module. Be strict on closed-notebook BUT generous on the open-response question — student paraphrases of the algebraic identity are valid even if not perfectly phrased.
- Project presentations: keep the energy up. Use stickers / acknowledgment for every pair. The presentation is partly a performance; affirm the work.
- If a pair didn’t finish their artefact, still let them present. The presentation itself is graded; the artefact is graded separately on the rubric.
- File the quiz results AND the project rubric scores to inform pacing for next year’s cohort.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2).